

标题

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6. Imperative paradigm: Programs consist of a sequence of commands.

Object-oriented paradigm: Programs are organized as active elements called objects.

Functional paradigm: Programs consist of nested functions.

Declarative paradigm: Programs consist of declarative statements that describe properties.

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23. Summarize the following rat's-nest routine with a single **if-else** statement:

```
if X > 5 then goto 80
```

```
X = X + 1
```

```
goto 90
```

```
80  X = X + 2
```

```
90  stop
```

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23. Summarize the following rat's-nest routine with a single `if-else` statement:

```
if X > 5 then goto 80
X = X + 1
goto 90
80  X = X + 2
90  stop
```

```
23.  if (X > 5):
        X = X + 2
    else:
        X = X + 1
```

25. Summarize the distinction between a translator and an interpreter.

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25. A translator merely converts a program from one language to another. An interpreter executes the source program without producing a translated copy.

32. What is the main difference between passing parameters to a function by value and passing parameters to a function by address/reference?

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32. a. Passing parameters by value protects the calling environment from being altered by the function being called.
- b. Passing parameters by reference is more efficient than passing them by value.